

**CONSTRUCTION OF AUTO-POWER SUPPLY FROM
FOUR SOURCES: MAINS, SOLAR, INVERTER AND
GENERATOR.**

BY

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**IN PARTIAL FULFILMENT OF THE AWARD OF BACHELOR OF
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ABSTRACT

Power is most important requirement for civilization. We know that because of large demand of electricity and due to limit capacity of power plant at generating station, power cut off is common for all us. Owing to limited amount of power generation at power station and due to shortage of non-renewable continuous supply source its beings a biggest challenge in Nigeria. If you see all around us you will observe that due to discontinuity of power supply, many problems have been faced by people in their everyday use. This type of power failure create problem for data centers, hospitals and some research work. This is a biggest reason that every country is researching for the work to supply a continuous power with good efficiency and with good regulation. In this project we can combine the renewable and non- renewable energy sources to get the continuous power supply such as mains, solar, inverter, petrol or diesel generator. The power cut of this source can be manually done by switches.

The continuous supply to load can be given by automatic operation of relay, relay driver IC with the help of microcontroller.

The main purpose of this project is to provide continuous power supply to a load, by selecting the supply from any of the four sources namely Mains, solar, inverter, and generator automatically in case if one the source is absent. The need of electricity is increasing day by day and the frequent power cuts of electricity are causing many problems in different areas like banks, colleges/schools, hospitals, houses and industries. Thus, there is requirement for an alternate arrangement of power supply.

When a source, say Mains fails, the supply automatically shifts to next priority source. On failure of the Mains supply the load gets supply from the next available source, say an inverter. If the inverter also fails it switches over to the next available source and so on. As it is not feasible to provide all 3 different sources of supply, one source with alternate switches

are provided to get the same function. The project can be further enhanced by using other sources like wind power also and then taking into consideration for using the best possible power whose tariff remains lowest at that moment.

DECLARATION

I hereby declare that I carried out the work reported in this report in the Department of Electrical and Computer Engineering Kwara State University Malete under the supervision of Mr. Tahir Aduragba. I also declare that to the best of my knowledge; no part of this project has been submitted elsewhere in a previous application for award of a degree. All sources of knowledge used have been duly acknowledged.

A handwritten signature in black ink, appearing to read 'Adewusi Oluwole Adewumi', is written over a horizontal dashed line.

Adewusi Oluwole Adewumi
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APPROVAL PAGE

This is to certify that the thesis titled ‘Auto-Power supply from 4 sources; Mains, Solar, Inverter and Generator’ by ADEWUSI OLUWOLE ADEWUMI with matriculation number 17D/7HEE/091 meets the requirements and regulations governing the award of the Bachelor of Engineering degree (B.Eng.) in Electrical and Computer Engineering of the Kwara State University, Malete, Kwara state, Nigeria.



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TABLE OF CONTENTS

	Page
TITLE PAGE	i
DEDICATION	ii
ABSTRACT	iii
ACKNOWLEDGEMENTS	v
DECLARATION	vi
APPROVAL PAGE	vii
TABLE OF CONTENTS	viii
LIST OF FIGURES	x
 CHAPTER	
1 INTRODUCTION	1
1.1 Background Study	1
1.2 Statement Problem	3
1.3 Aims and Objective	3
1.4 Scope of the Project	4
1.5 Chapter Breakdown	5
2 LITERATURE REVIEW	6
2.1 Overview of power supply unit	6
2.2 Review of the four different supplies	7
2.2.1 Mains Electricity Supply	7
2.2.2 Solar Power	10
2.2.3 Inverter	12
2.2.4 Generator	14
2.3 History of Power Failure and System Reliability	16
2.4 Types of Power Failure	17
2.5 Power Failure Inevitability and Electric Sustainability	18
3 DESIGN AND IMPLEMENTATION	21
3.1 Device Block Diagram	21
3.2 Circuit Schematic	22
3.3 Hardware Requirements	23

3.4 Software Requirements	23
3.5 Review of System Components	24
3.6 Device Working Principle	25
3.7 Flowchart of the Circuit	27
4 TEST RESULTS AND DISCUSSION	28
4.1 Construction of the Casing and Packaging	28
4.2 Economics and Project Viability	29
4.3 Reliability and Maintainability of the device	31
4.4 Project Evaluation and Testing	32
4.5 Project Application	32
5 CONCLUSIONS AND RECOMMENDATIONS	33
5.1 Conclusions	33
5.2 Future Scope and Recommendations	33
REFERENCES	34
APPENDICES	35

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND STUDY

The auto-power supply control system is a cost effective and convenient system for that consumer who want to attain uninterrupted power supply from sources such as Main, Solar, Inverter and Generator. Looking from a commercial level, we can estimate that there are so many consumers that have the equipment or machines whose requirement is based on an uninterrupted power supply. For example, a bank or a hospital who is in need of an uninterrupted power supply because damage could have been done from manually switching to other power source. From the above examples, we can base the importance of the auto power supply control system. There are some consumers that switches based on manual control system which are making use of a magnetic contactors and power relays but their system is not effective and do not provide precise uninterruptible power supply. Here, we are making this auto-power supply control system with the help of power electronics components, microcontroller and electronic relays.

An important requirement of electric power distribution systems is the need for automatic operation. In particular, the rapid and reliable transfer of the system from one power source to another during certain system events is important to achieving the reliability goals for such systems and the facility serves. However, the design of such an automatic transfer system is all too-often considered “less important” than many other aspects of the overall power system design. Electricity has become indispensable to socio-economic and industrial development of any nation. Using uninterrupted power supply in an automated mode, we always have a substitute arrangement as backup to take place of main power supply in case of power-cut in an emergency case, where the power cut cannot be avoided.

For daily usage, voltage supplied is within 240V AC. The need for access to electricity as the basic form of energy supply, uninterrupted power supply to the masses is paramount for the growth of a country and vital for the development of a nation's economy. The strategic role and policy of uninterrupted supply of electricity in the development of an economy has always been appreciated by most developed nations, with the likes of France, Germany, and Italy. The power sector provides a platform for economic development; electricity has brought about development in all area of productions and services. Electricity has become indispensable to socio-economic and industrial development of any nation.

Using uninterrupted power supply in an automated mode, is the best that can happen to a developing country like Nigeria and its importance cannot be overemphasized.

A power supply in an electronic device supplies electric energy to an electrical load. The primary function of a power supply is to convert one form of electrical energy to another. As a result, power supplies are sometimes referred to as electric power converters. Some power supplies are discrete, stand-alone devices, whereas others are built into larger devices along with their loads. Examples of the latter include power supplies found in desktop computers and consumer electronics devices.

Every power supply must obtain the energy it supplies to its load, as well as any energy it consumes while performing that task, from an energy source. Depending on its design, a power supply may obtain energy from various types of energy sources, including electrical energy transmission systems, energy storage devices such as a batteries and fuel cells, electromechanical systems such as generators and alternators, solar power converters, or another power supply.

In this project, four switches were used to demonstrate the respective failure of that power supply. When any of the switches is pressed it shows the absence of that particular source, switches are connected to microcontroller as input signals. A PIC microcontroller 16F8768A is used.

The output of microcontroller is given to the relay driver IC, which switches appropriate relay to maintain uninterrupted supply to the load. The output shall be observed using a lamp drawing power supply from mains supply initially. On failure of the mains supply (which is actuated by pressing the appropriate switch) the load gets supply from the next available source, say Solar.

If the next supply also fails it switches over to the next available source and so on. The current status, as to which source supplies the load is also displayed on an LCD. As it is not feasible to provide all 4 different sources of supply, one source with alternate switches are provided to get the same function.

1.2 STATEMENT PROBLEM

The absence or epileptic provision of power supply has caused a lot of issue in a nation like Nigeria, especially in economic terms.

This project is aims to mitigate this problem by providing alternative sources. Also, the alternative sources are provided uninterruptedly.

In Nigeria, stable power supply has been a major concern. Nowadays, as the demand for electricity is increasing every day and frequent power outage is causing many problems in areas like industries, banks, hospitals and houses. An alternative arrangement for power source is a must.

To eliminate the problem of power outages and data loss by maintaining power to the critical loads.

1.3 AIMS AND OBJECTIVES

Thus, the objectives from the problem statement are:

1. To design a device that will provide uninterrupted power supply to a load, by selecting the supply source automatically from any available one out of 3 such as: Solar, Inverter or Generator in the absence of Main power supply. This is done by using microcontroller, relay, switches and LCD.
2. To protect the critical loads from power interruptions.
3. To analyze a complete circuit design.
4. To make comparison between simulation & experimental result.

At the end of this work, I will be able to establish:

- i. Operation and description of the four different supply.
- ii. Operation of relays and their connections
- iii. How to connect switches, and LCD

1.4 SCOPE OF THE PROJECT

This project entails the following:

1. Developing a circuitry that shows how to implement an Auto-power supply circuit.
2. Designing a circuit that incorporates all the key components required in the design of an Auto-power supply circuit.
3. Comparison between results of experimental & hardware is used for analysis & verification.

Four switches are used for four respective sources; the main supply, the solar supply, the inverter and generator. These are connected to a microcontroller of 8051 family that provides input signals to it. Whenever a switch is pressed it shows the absence of that particular source. A relay driver is used that receives microcontroller generated output and switches that particular relay to provide continuous power supply. A lamp is used as a load for demonstration purpose which draws power from main supply from the grid. When main supply fails to supply power, automatically next available source is used say Solar. If Solar fails then the next one is used and so on. An LCD is also used to display which source is being currently used for power supply.

1.5 CHAPTER BREAKDOWN

The various stages involved in the development of this project have been properly put into five chapters to enhance comprehensive and concise reading. In this project thesis, the chapter is organized sequentially as follows:

Chapter Two: is on literature review of this study. In this chapter, all the literature pertaining to this work was reviewed.

Chapter Three: is on design methodology. In this chapter all the method involved during the design and construction were discussed.

Chapter Four: is on Reliability and testing analysis. All testing that result accurate functionality was analysed.

Chapter Five: is on Conclusion, Recommendation and References.

CHAPTER TWO

LITERATURE REVIEW

Power failure is one of the major problems Nigeria is facing. Some important literatures are review as below:

2.1 OVERVIEW OF POWER SUPPLY UNIT

A power supply unit is an electrical device that supplies electric power to an electrical load. The primary function of a power supply is to convert electric current from a source to the correct voltage, current, and frequency to power the load. As a result, power supplies are sometimes referred to as electric power converters. Some power supplies are separate standalone pieces of equipment, while others are built into the load appliances that they power. Examples of the latter include power supplies found in desktop computers and consumer electronics devices. Other functions that power supplies may perform include limiting the current drawn by the load to safe levels, shutting off the current in the event of an electrical fault, power conditioning to prevent electronic noise or voltage surges on the input from reaching the load, power-factor correction, and storing energy so it can continue to power the load in the event of a temporary interruption in the source power (uninterruptible power supply).

All power supplies have a power input connection, which receives energy in the form of electric current from a source, and one or more power output connections that deliver current to the load. The source power may come from the electric power grid, such as an electrical outlet, energy storage devices such as batteries or fuel cells, generators or alternators, solar power converters, or another power supply. The input and output are usually hardwired circuit connections, though some power supplies employ wireless energy transfer to power their loads without wired

connections. Some power supplies have other types of inputs and outputs as well, for functions such as external monitoring and control.

2.2 REVIEW OF THE FOUR DIFFERENT SUPPLIES

The four different type of power supply used in this project is reviewed below:

2.2.1 MAINS ELECTRICITY SUPPLY

Mains electricity is the general-purpose alternating-current (AC) electric power supply which comes from the grid. Mains electricity is the form of electrical power that is delivered to homes and businesses. It is the form of electrical power that consumers use when they plug kitchen appliances, televisions and electric lamps into wall sockets.

The two principal properties of the electric power supply, voltage and frequency, differ between regions. A voltage of (nominally) 220 V and a frequency of 50 Hz is used in Europe, most of Africa, most of Asia, much of South America and Australia. In North America, the most common combination is 120 V and a frequency of 60 Hz. Other voltages exist, and some countries may have, for example, 230 V but 60 Hz. This is a concern to travelers, since portable appliances designed for one voltage and frequency combination may not operate with, or may even be destroyed by another. The use of different and incompatible plugs and sockets in different regions and countries provides some protection from accidental use of appliances with incompatible voltage and frequency requirements.

In 1881 two electricians built the world's first power system at Godalming in England. It was powered by a power station consisting of two waterwheels that produced an alternating current that in turn supplied seven Siemens arc lamps at 250 volts and 34 incandescent lamps at 40 volts.^[1] However supply to the lamps was intermittent and in 1882 Thomas Edison and his

company, The Edison Electric Light Company, developed the first steam powered electric power station on Pearl Street in New York City. The Pearl Street Station initially powered around 3,000 lamps for 59 customers.^{[2][3]} The power station used direct current and operated at a single voltage. Direct current power could not be easily transformed to the higher voltages necessary to minimize power loss during long-distance transmission, so the maximum economic distance between the generators and load was limited to around half-a-mile (800 m).^[4]

That same year in London Lucien Gaulard and John Dixon Gibbs demonstrated the first transformer suitable for use in a real power system. The practical value of Gaulard and Gibbs' transformer was demonstrated in 1884 at Turin where the transformer was used to light up forty kilometers (25 miles) of railway from a single alternating current generator.^[5] Despite the success of the system, the pair made some fundamental mistakes. Perhaps the most serious was connecting the primaries of the transformers in series so that active lamps would affect the brightness of other lamps further down the line.

In 1885 George Westinghouse, an American entrepreneur, obtained the patent rights to the Gaulard Gibbs transformer and imported a number of them along with a Siemens generator and set his engineers to experimenting with them in the hopes of improving them for use in a commercial power system. One of Westinghouse's engineers, William Stanley, recognized the problem with connecting transformers in series as opposed to parallel and also realized that making the iron core of a transformer a fully enclosed loop would improve the voltage regulation of the secondary winding. Using this knowledge, he built the first practical transformer based alternating current power system at Great Barrington, Massachusetts in 1886.^[6] Westinghouse would begin installing multi-voltage AC transformer systems in competition with the Edison company later that year. In 1888 Westinghouse also licensed Nikola Tesla's US patents for a

polyphase AC induction motor and transformer designs and hired Tesla for one year to be a consultant at the Westinghouse Electric & Manufacturing Company's Pittsburgh labs.^[7]

By 1888 the electric power industry was flourishing, and power companies had built thousands of power systems (both direct and alternating current) in the United States and Europe. These networks were effectively dedicated to providing electric lighting. During this time the rivalry between Thomas Edison and George Westinghouse's companies had grown into a propaganda campaign over which form of transmission (direct or alternating current) was superior, a series of events known as the "War of Currents".^[8] In 1891, Westinghouse installed the first major power system that was designed to drive a 100 horsepower (75 kW) synchronous electric motor, not just provide electric lighting, at Telluride, Colorado.^[9] On the other side of the Atlantic, Mikhail Dolivo-Dobrovolsky built a 20 kV 176 km three-phase transmission line from Lauffen am Neckar to Frankfurt am Main for the Electrical Engineering Exhibition in Frankfurt.^[10] In the US the AC/DC competition came to the end when Edison General Electric was taken over by their chief AC rival, the Thomson-Houston Electric Company, forming General Electric. In 1895, after a protracted decision-making process, alternating current was chosen as the transmission standard with Westinghouse building the Adams No. 1 generating station at Niagara Falls and General Electric building the three-phase alternating current power system to supply Buffalo at 11 kV.

Developments in power systems continued beyond the nineteenth century. In 1936 the first experimental HVDC (high voltage direct current) line using mercury arc valves was built between Schenectady and Mechanicville, New York. HVDC had previously been achieved by series-connected direct current generators and motors (the Thury system) although this suffered from serious reliability issues.^[11] In 1957 Siemens demonstrated the first solid-state rectifier, but it was not until the early 1970s that solid-state devices became the standard in HVDC.^[12]

In recent times, many important developments have come from extending innovations in the ICT field to the power engineering field. For example, the development of computers meant load flow studies could be run more efficiently allowing for much better planning of power systems. Advances in information technology and telecommunication also allowed for remote control of a power system's switchgear and generators.

2.2.2 SOLAR POWER

Solar power is the conversion of energy from sunlight into electricity, either directly using photovoltaics (PV), indirectly using concentrated solar power, or a combination. Concentrated solar power systems use lenses or mirrors and tracking systems to focus a large area of sunlight into a small beam. Photovoltaic cells convert light into an electric current using the photovoltaic effect.^[19]

Photovoltaics were initially solely used as a source of electricity for small and medium-sized applications, from the calculator powered by a single solar cell to remote homes powered by an off-grid rooftop PV system. Commercial concentrated solar power plants were first developed in the 1980s. The 392 MW Ivanpah installation is the largest concentrating solar power plant in the world, located in the Mojave Desert of California.

As the cost of solar electricity has fallen, the number of grid-connected solar PV systems has grown into the millions and utility-scale solar power stations with hundreds of megawatts are being built. Solar PV is rapidly becoming an inexpensive, low-carbon technology to harness renewable energy from the Sun. The current largest photovoltaic power station in the world is the 850 MW Longyangxia Dam Solar Park, in Qinghai, China.

The International Energy Agency projected in 2014 that under its "high renewables" scenario, by 2050, solar photovoltaics and concentrated solar power would contribute about 16 and 11 percent, respectively, of the worldwide electricity consumption, and solar would be the world's largest source of electricity. Most solar installations would be in China and India.^[20] Currently, as of 2016, solar power provides just 1% of total worldwide electricity production but growing 33% per annum.

The early development of solar technologies starting in the 1860s was driven by an expectation that coal would soon become scarce. Charles Fritts installed the world's first rooftop photovoltaic solar array, using 1%-efficient selenium cells, on a New York City roof in 1884.^[21] However, development of solar technologies stagnated in the early 20th century in the face of the increasing availability, economy, and utility of coal and petroleum.^[22] In 1974 it was estimated that only six private homes in all of North America were entirely heated or cooled by functional solar power systems.^[23] The 1973 oil embargo and 1979 energy crisis caused a reorganization of energy policies around the world and brought renewed attention to developing solar technologies.^{[24][25]} Deployment strategies focused on incentive programs such as the Federal Photovoltaic Utilization Program in the US and the Sunshine Program in Japan. Other efforts included the formation of research facilities in the United States (SERI, now NREL), Japan (NEDO), and Germany (Fraunhofer-ISE).^[26] Between 1970 and 1983 installations of photovoltaic systems grew rapidly, but falling oil prices in the early 1980s moderated the growth of photovoltaics from 1984 to 1996.

In the mid-1990s, development of both, residential and commercial rooftop solar as well as utility-scale photovoltaic power stations, began to accelerate again due to supply issues with oil and natural gas, global warming concerns, and the improving economic position of PV relative to other energy technologies.^[27] In the early 2000s, the adoption of feed-in tariffs—a policy

mechanism, that gives renewables priority on the grid and defines a fixed price for the generated electricity—lead to a high level of investment security and to a soaring number of PV deployments in Europe.

2.2.3 INVERTER

Inverter is an electronic circuitry that changes direct current (DC) to alternating current (AC).^[15] The input voltage, output voltage and frequency, and overall power handling depend on the design of the specific device or circuitry. The inverter does not produce any power on its own; the power is provided by the DC source.

A typical power inverter device or circuit requires a relatively stable DC power source capable of supplying enough current for the intended power demands of the system. The input voltage depends on the design and purpose of the inverter. Examples include:

- 12 V DC, for smaller consumer and commercial inverters that typically run from a rechargeable 12 V lead acid battery or automotive electrical outlet.^[16]
- 24, 36 and 48 V DC, which are common standards for home energy systems.
- 200 to 400 V DC, when power is from photovoltaic solar panels.
- 300 to 450 V DC, when power is from electric vehicle battery packs in vehicle-to-grid systems.
- Hundreds of thousands of volts, where the inverter is part of a high-voltage direct current power transmission system.

An inverter can produce a square wave, modified sine wave, pulsed sine wave, pulse width modulated wave (PWM) or sine wave depending on circuit design. The two dominant commercialized waveform types of inverters as of 2007 are modified sine wave and sine wave.

There are two basic designs for producing household plug-in voltage from a lower-voltage DC source, the first of which uses a switching boost converter to produce a higher-voltage DC and then converts to AC. The second method converts DC to AC at battery level and uses a line-frequency transformer to create the output voltage.^[17]

From the late nineteenth century through the middle of the twentieth century, DC-to-AC power conversion was accomplished using rotary converters or motor-generator sets (M-G sets). In the early twentieth century, vacuum tubes and gas filled tubes began to be used as switches in inverter circuits. The most widely used type of tube was the thyratron.

The origins of electromechanical inverters explain the source of the term inverter. Early AC-to-DC converters used an induction or synchronous AC motor direct-connected to a generator (dynamo) so that the generator's commutator reversed its connections at exactly the right moments to produce DC. A later development is the synchronous converter, in which the motor and generator windings are combined into one armature, with slip rings at one end and a commutator at the other and only one field frame. The result with either is AC-in, DC-out. With an M-G set, the DC can be considered to be separately generated from the AC; with a synchronous converter, in a certain sense it can be considered to be "mechanically rectified AC". Given the right auxiliary and control equipment, an M-G set or rotary converter can be "run backwards", converting DC to AC. Hence an inverter is an inverted converter.^[18]

2.2.4. GENERATOR

A generator is the combination of a engine with an electric generator (often an alternator) to generate electrical energy. This is a specific case of engine-generator. A compression-ignition engine is usually designed to run on diesel fuel, but some types are adapted for other liquid fuels or natural gas.

Diesel generating sets are used in places without connection to a power grid, or as emergency power-supply if the grid fails, as well as for more complex applications such as peak-opping, grid support and export to the power grid.

The packaged combination of a diesel engine, a generator and various ancillary devices (such as base, canopy, sound attenuation, control systems, circuit breakers, jacket water heaters and starting system) is referred to as a "generating set" or a "genset" for short.

Set sizes range from 8 to 30 kW (also 8 to 30 kVA single phase) for homes, small shops and offices with the larger industrial generators from 8 kW (11 kVA) up to 2,000 kW (2,500 kVA three phase) used for large office complexes, factories. A 2,000-kW set can be housed in a 40 ft (12 m) ISO container with fuel tank, controls, power distribution equipment and all other equipment needed to operate as a standalone power station or as a standby backup to grid power. These units, referred to as power modules are gensets on large triple axel trailers weighing 85,000 pounds (38,555 kg) or more. A combination of these modules is used for small power stations and these may use from one to 20 units per power section and these sections can be combined to involve hundreds of power modules. In these larger sizes the power module (engine and generator) are brought to site on trailers separately and are connected together with large cables and a control cable to form a complete synchronized power plant. A number of options also exist to tailor specific needs, including control panels for auto start and mains paralleling, acoustic canopies for fixed or mobile applications, ventilation equipment, fuel supply systems, exhaust systems, etc.^[1] Diesel generators, sometimes as small as 200 kW (250 kVA) are widely used not only for emergency power, but also many have a secondary function of feeding power to utility grids either during peak periods, or periods when there is a shortage of large power generators.

Ships often also employ diesel generators, sometimes not only to provide auxiliary power for lights, fans, winches etc., but also indirectly for main propulsion. With electric propulsion the generators can be placed in a convenient position, to allow more cargo to be carried. Electric drives for ships were developed prior to World War I. Electric drives were specified in many warships built during World War II because manufacturing capacity for large reduction gears was in short supply, compared to capacity for manufacture of electrical equipment.^[2] Such a diesel-electric arrangement is also used in some very large land vehicles such as railroad locomotives.

Generating sets are selected based on the electrical load they are intended to supply, the electrical load's characteristics such as kW, kVA, var, harmonic content, surge currents (e.g., motor starting current) and non-linear loads. The expected duty (such as emergency, prime or continuous power) as well as environmental conditions (such as altitude, temperature and exhaust emissions regulations) must also be considered.

Proper sizing of diesel generators is critical to avoid low-load or a shortage of power. Sizing is complicated by the characteristics of modern electronics, specifically non-linear loads. In size ranges around 50 MW and above, an open cycle gas turbine is more efficient at full load than an array of diesel engines, and far more compact, with comparable capital costs; but for regular part-loading, even at these power levels, diesel arrays are sometimes preferred to open cycle gas turbines, due to their superior efficiencies.

2.3 HISTORY OF POWER FAILURE AND SYSTEM RELIABILITY

On August 14, 2003, the most catastrophic, widespread failure of the US power grid occurred resulting in the loss of electrical service for approximately 55 million people and lasted as long as two days in some locations. The incident contributed to at least eleven fatalities and a financial

loss of over \$6 billion dollars. The power failures began in Northern Ohio, where a fault in a single power line caused a series of cascading failures that affected the Cleveland area, the entire state of Michigan, and large portions of Pennsylvania, New York, New Jersey, Massachusetts, Connecticut, and South-eastern Canada. This was one of the worst power outages at the time, second only to the 1999 Southern Brazil blackout.

Most people take electricity and the electrical systems the power serves for granted; never really thinking about it until the power goes out leaving them feeling helpless. During an outage, some may find themselves stuck in an elevator or subway. Without power, traffic lights will not function causing traffic jams and automobile accidents. Water supplies shut down because the pumps cannot work without power. Food spoils due to failed refrigeration systems. Health and emergency services become compromised and the risks to public safety and security greatly increase. Every aspect of our high-tech systems would be compromised and fire protection and life safety systems are no exception, which was evident during the Northeast blackout of 2003. The Northeast blackout of 2003 crippled telephone systems, cell-phone service, and transportation systems, which were all dependent upon the electrical power network. Not only did it affect civilian and private systems, there was also a failure of major emergency communications systems including 911 systems, and FDNY dispatchers. Fire alarm systems powered by standby battery back-up typically failed after 24 hours. In fact, the only systems that did not fail were the “vintage” emergency public telegraph systems, many of which are still in service in several municipalities in the Northeast Region of the United States. Today, most telegraph systems have been decommissioned throughout the country primarily because they have been replaced by more modern fire department dispatch systems such as 911 systems and private cellular networks. Yet as this historic failure has shown us, the functionality

of these older systems is worth taking a closer look at because these were the only emergency systems that operated properly during this blackout

2.4 TYPES OF POWER FAILURE

Power outages are categorized into three different phenomena, relating to the duration and effect of the outage:

- A transient fault is a momentary (a few seconds) loss of power typically caused by a temporary fault on a power line. Power is automatically restored once the fault is cleared.
- A brownout or sag is a drop-in voltage in an electrical power supply. The term brownout comes from the dimming experienced by lighting when the voltage sags. Brownouts can cause poor performance of equipment or even incorrect operation.
- A blackout refers to the total loss of power to an area and is the most severe form of power outage that can occur. Blackouts which result from or result in power stations tripping are particularly difficult to recover from quickly. Outages may last from a few minutes to a few weeks depending on the nature of the blackout and the configuration of the electrical network.

Restoring power after a wide-area outage can be difficult, as power stations need to be brought back on-line. Normally, this is done with the help of power from the rest of the grid. In the total absence of grid power, a so-called black start needs to be performed to bootstrap the power grid into operation. The means of doing so will depend greatly on local circumstances and operational policies, but typically transmission utilities will establish localized 'power islands' which are then progressively coupled together. To maintain supply frequencies within tolerable limits during this process, demand must be reconnected at the same pace that generation is restored, requiring close coordination between power stations, transmission and distribution organizations.

2.5 POWER FAILURE INEVITABILITY AND ELECTRIC SUSTAINABILITY

It has been argued on the basis of historical data and computer modeling that power grids are self-organized critical systems. These systems exhibit unavoidable disturbances of all sizes, up to the size of the entire system. This phenomenon has been attributed to steadily increasing demand/load, the economics of running a power company, and the limits of modern engineering. While blackout frequency has been shown to be reduced by operating it further from its critical point, it generally isn't economically feasible, causing providers to increase the average load over time or upgrade less often resulting in the grid moving itself closer to its critical point. Conversely, a system past the critical point will experience too many blackouts leading to system-wide upgrades moving it back below the critical point. The term critical point of the system is used here in the sense of statistical physics and nonlinear dynamics, representing the point where a system undergoes a phase transition; in this case the transition from a steady reliable grid with few cascading failures to a very sporadic unreliable grid with common cascading failures. Near the critical point the relationship between blackout frequency and size follows a power law distribution. Other leaders are dismissive of system theories that conclude that blackouts are inevitable, but do agree that the basic operation of the grid must be changed. The Electric Power Research Institute champions the use of smart grid features such as power control devices employing advanced sensors to coordinate the grid. Others advocate greater use of electronically controlled High-voltage direct current (HVDC) firebreaks to prevent disturbances from cascading across AC lines in a wide area grid.

Cascading failure becomes much more common close to this critical point. point, these failures have a greater effect on the surrounding components due to individual components carrying a larger load. This results in the larger load from the failing component having to be redistributed in larger quantities across the system, making it more likely for additional components not

directly affected by the disturbance to fail, igniting costly and dangerous cascading failures. These initial disturbances causing blackouts are all the more unexpected and unavoidable due to actions of the power suppliers to prevent obvious disturbances (cutting back trees, separating lines in windy areas, replacing aging components etc.). The complexity of most power grids often makes the initial cause of a blackout extremely hard to identify. The power law relationship is seen in both historical data and model systems. The practice of operating these systems much closer to their maximum capacity leads to magnified effects of random, unavoidable disturbances due to aging, weather, human interaction etc.

CHAPTER THREE

DESIGN AND IMPLEMENTATION

3.1 DEVICE BLOCK DIAGRAM

Before carrying out any project, the block diagram must be drawn and fully understood. Block diagram gives a pictorial understanding of any work. Here is the block diagram of the auto power supply control system from 4 different sources using **pic microcontroller** with all the essential components. The Block Diagram of the auto power supply control system from 4 different sources using pic microcontroller:

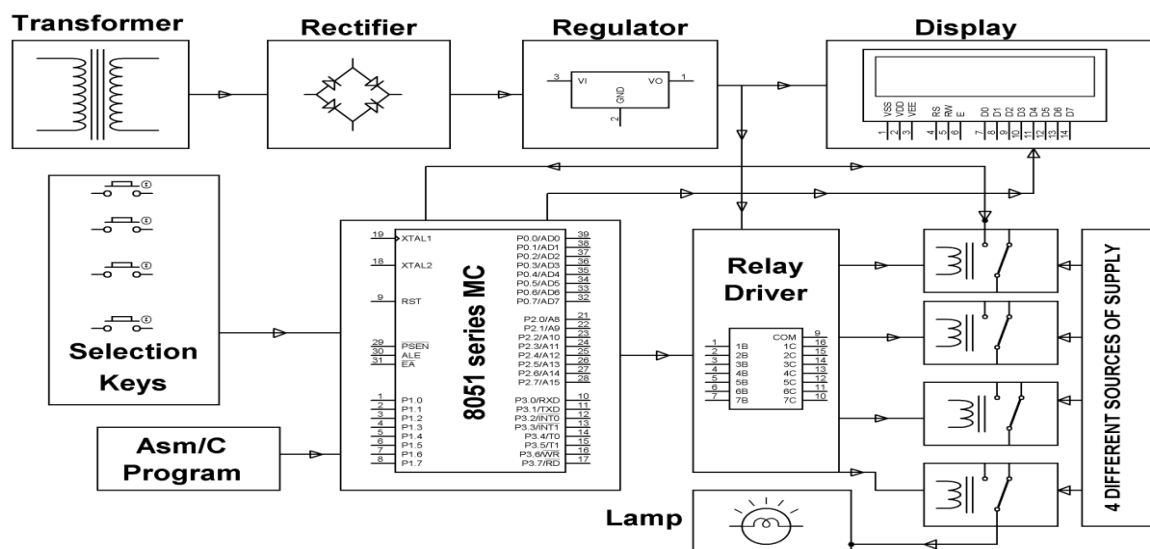


Figure 3.1: Block Diagram of the Device

3.2 CIRCUIT SCHEMATIC

The circuit diagram of the system is as below:

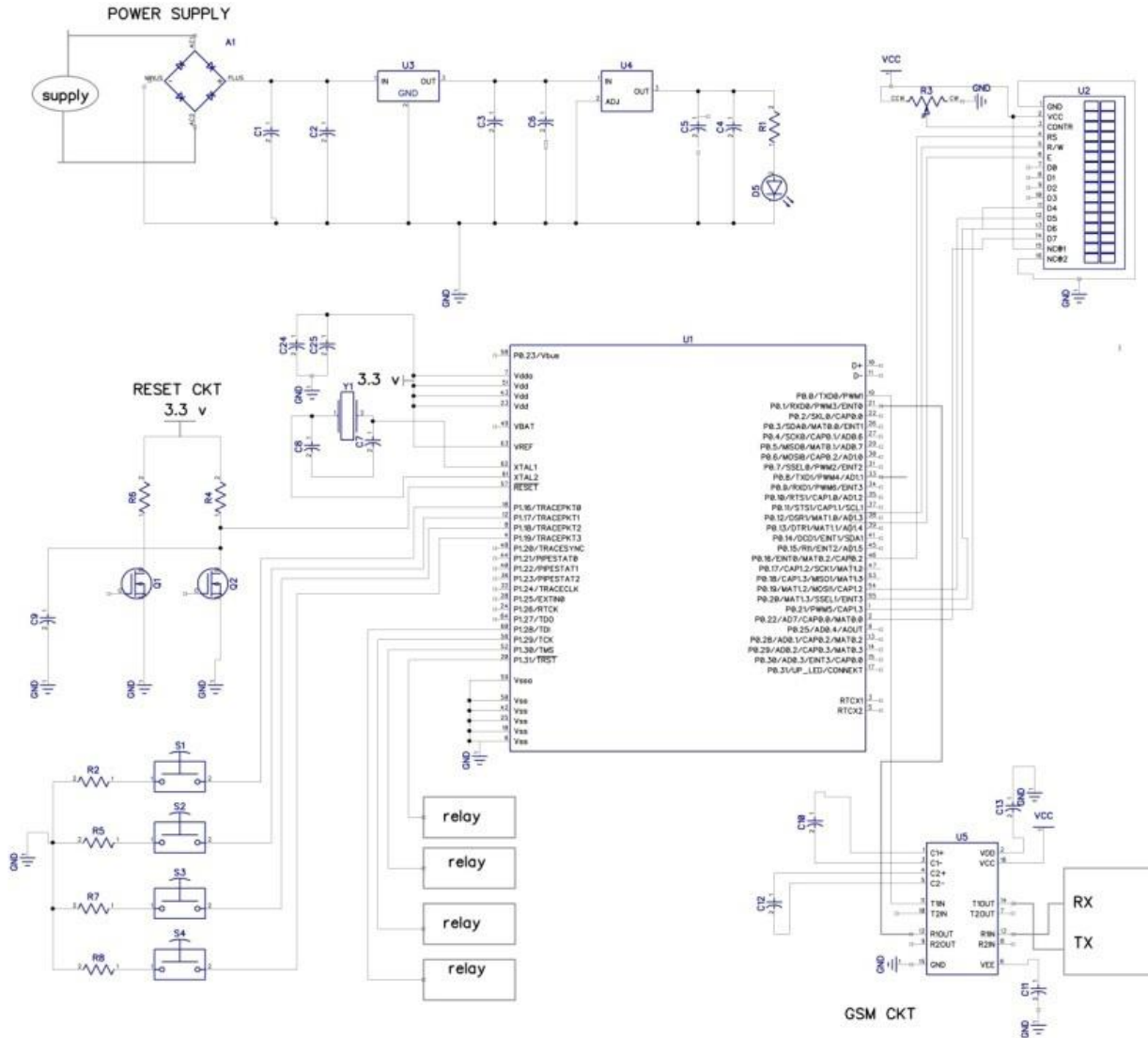


Figure 3.2: Schematic Diagram of the circuit

3.3 HARDWARE REQUIREMENTS

- PIC Microcontroller 16F876A
- Relay Driver IC ULN 2003
- Relays
- LCD
- Lamp
- Toggle switch
- Transformer
- Diodes
- Voltage Regulator
- LEDS
- Capacitors
- Resistors

3.4 SOFTWARE REQUIREMENTS

- MikroBasic compiler

Language: Basic Language

3.5 REVIEWS OF SYSTEM COMPONENTS

Microcontroller PIC 16F876A: In this auto power supply control system, the PIC 18F876A microcontroller is used for the auto selection of the available source. It shifted the load to the other power supply source automatically without any interruption. It is programmed in C language with help of micro c software and is powered up with 5V dc voltages. It is interfaced with LCD display and relay driver IC.

Relay Driver: In this auto power supply control system, the relay driver ULN2003 is used for driving the load relays. This relay receives the signal from microcontroller for shifting the load on another supply source. It is powered up with 5V dc and interfaced with microcontroller.

Transformer: The transformer is used for connecting this system directly to 220V AC. It steps down 220V into 12V.

Voltage Regulator: The voltage regulator is used for regulating 12V DC into 5V DC voltages for supplying the power to the LED, microcontroller and other components. IC LM 7805 voltage regulator is used for regulating voltages.

LCD Display: LCD display is used for displaying the source of supply on which the whole system is working. The LCD used in this project is a 16x2 LCD.

Capacitor: Capacitive filters are used in this work. It removes the ripples from output of the rectifier. The DC output received from this filter is constant until the mains voltage and load is maintained constant but if either one of the quantities is vary; the DC output received at this point changes. To overcome this drawback a regulator is applied at the output stage of the filter.

Bridge Rectifier: In this work bridge rectifier is used due to its merits like full wave rectification and high stability. For a single half of the cycle only two diodes will be in forward bias condition.

Relay: Relays are electromechanical devices or solid-state devices which operate in response to a signal which may be voltage, current, temperature etc. Electromagnetic relays operate due to magnetic fields. They are composed basically of two parts: (1) The operating coil and (2) The magnetic switch. When an input pulse is introduced into the coil, a magnetic field is produced in the core of the electromagnet. This action causes the switch to slide. Relays are either normally open or normally close. Relays are available for DC or AC excitation and coil voltages range from 5V to 230V.

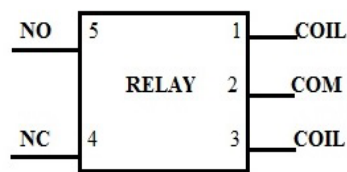


Figure 3.3: Pin Diagram

Toggle switch. A toggle switch is actuated by moving a lever back and forth to open or close an electrical circuit.

3.6 DEVICE WORKING PRINCIPLE

This device uses an arrangement of four different sources of supply which are channelized to a load so as to have an uninterrupted operation of the load. As it is not practicable to get four sources of supply running at the same time to the load, I made use of one source and a set of relays. I took first source with main supply and assumed it is being fed from four different sources by connecting all the four incoming sources in parallel. The AC source to the lamp is connected to four relays by making the entire normally open contacts parallel and all the common contacts in parallel. Four push button switches are used which represent failure of corresponding supply respectively and are interfaced to the controller.

Initially, I gave high input signal to the microcontroller, so as a result the controller generates a low output to activate the first relay driver which will result in the relay being energized and the lamp glows. While the push button for Mains is pressed that represents failure of main supply, as a result the supply is provided from the next source and the microcontroller receive high input and generates low output to activate the second relay driver which will result in the second relay being energized and the lamp glows. When the Solar button is pressed, the supply comes from the next source which will supply high input to the controller and which will provide low signal to the third relay and the lamp switches ON and when the third push button is pressed, the supply will chose next source now the fourth source, which will provide input to the microcontroller and controller activates the fourth relay and the load will get the supply and the lamp continues to glow. When all the relays are off leaving no supply to the lamp, the lamp is switched off.

One point to note is that the Mains is top priority, which means that at any instance, if the main supply is available, it overrides the sequence and the supply comes from the main supply.

One 16 x 2 lines LCD is used to display the condition of the supply sources and the load on real time basis.

3.7 FLOWCHART OF THE CIRCUIT

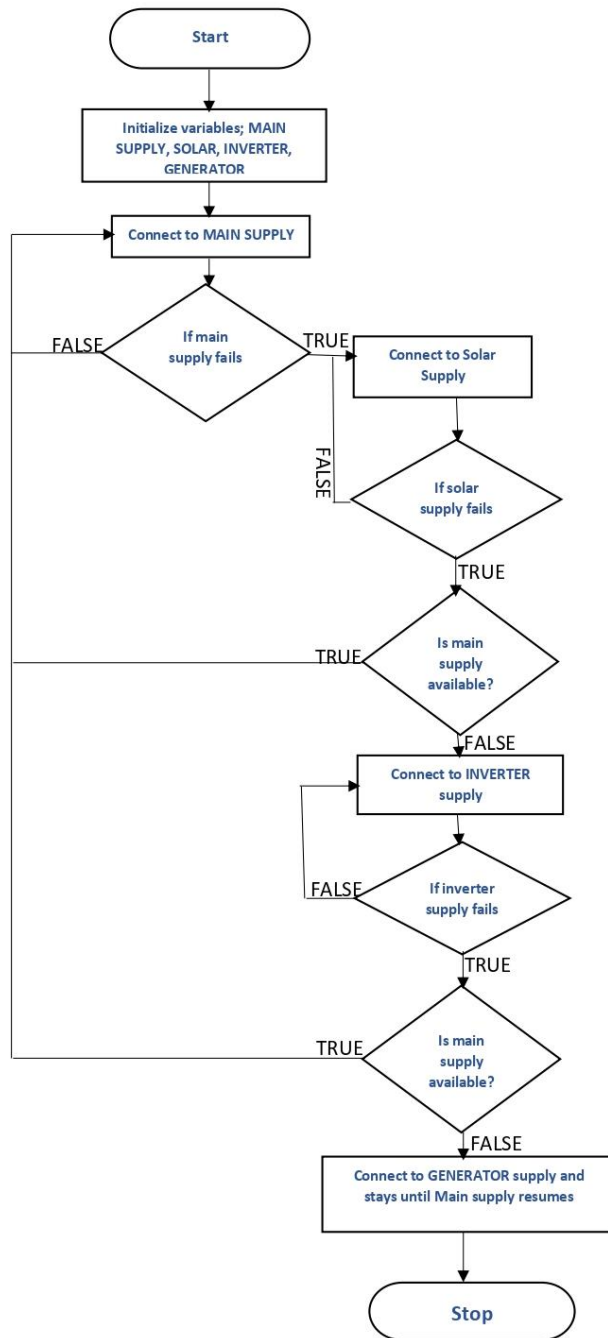


Figure 3.4: Flowchart of the circuit

CHAPTER FOUR

TESTS, RESULT AND DISCUSSIONS.

In building this project, the following procedures were properly considered,

- I. Proposing of the entire materials / Components needed
- ii. Resistance check of the components with the help of ohmmeter before making the necessary connection with the components
- iii. Drafting out a schematic diagram or how to arrange the materials / components.
- iv. Testing the completed system to see if the design works and
- v. Finally, implementation of design of the project.

Having procured all the materials, I processed into the arrangement of the components into the Vero board, proper soldering of the components then followed. The components were all soldered into the board after which it was correctly confirmed done.

4.1 CONSRUCTION OF THE CASING AND PACKAGING

The casing of this project comprises of internal and external packaging. Internal parts of the circuit were soldered and fixed inside the case. Then after that, follow by external components such as indicators, switches, and connectors.

Having provided the casing and having finished the construction of the sections of this system, the assembling into the casing followed. The sections were properly laid out and assembled into the casing where the general coupling and linkages into the peripheral devices took place.

Finally; the indicator, switches, connectors were carefully brought out from the internal part of the casing through the holes made on the body of the casing, the input cable plug outlet mounted on the body of the casing where power source terminals will be connected to.

After the procurement of the metal sheet and due consideration of the circuit arrangements, I was able to determine the size of the casing which measures 450mm by 300mm by 180mm and that of the mounting base.

4.2 ECONOMICS AND PROJECT VIABILITY

Although this project has not been given due recognition by the authority concerned, whenever this equipment finds its use the case is relatively cheap with a good efficiency and improves on its reliability. Due attention will be given to the viability of this project reliability maintainability and also the evaluation.

In the manufacturing of any technological advancement or master piece the designer is expected to evaluate to what extent the equipment will stand the test of time. It is on this note that I have dedicated the section to explain the viability of this project.

It has been observed that due to the carefree attitude of most operators at the distribution stations, power supplies to vast areas under control has witnessed unprecedented power interruption. This has caused untold hardship in mainly Hospital environments and in most cases frustrated the effort of some researchers in the laboratory countless numbers of information has gone unretrieved in computer centers. The idea of preserving food items in cold rooms have long been forgotten. This ugly trend cannot be allowed to divide the existing standard of the populace. It should be noted that there is no economically viable electrical equipment without any regulation guiding it. This is to clear the view held against this project that the proliferation of it the service area could lead to over loading of supply authority's equipment, unbalanced phase and consequent breakdown of equipment.

Since the majority of problems facing the supply authority is the strain on the service lines which results from addition of new heavy equipment that are not including in the original of the installation.

This project was not designed towards the effort of supply authority's but rather it mainly aimed at reducing the high risk involved in the manual changing of the supply in the event of power failure in the functional device. Since the arching of the contacts several times during reconnection or swapping causes high intermittent power loss across the contacts this in turn will result in over loading of the supply equipment. But with this device installed, just one-time arcing is sufficient to reconnect and disconnect the circuit thereby reducing the power loss. This equipment will be of great benefit to the following classes of consumers.

1. Hospital theatres
2. Conference halls
3. Computer centers
4. Monitoring centers
5. Data and research centers
6. Laboratories etc

4.3 RELIABILITY AND MAINTAINABILITY OF THE DEVICE

In the design of this device, reliability is taken into consideration to improve on the system performance. Here the concept of reliability has been associated, in a qualitative way with good design endurance consistence quality and dependability in recent years however, the much greater complexity of the line selector and the seriousness of a failure in the system have made it necessary to attempt not only to improve the reliability of the equipment but also to assesses it in qualitative terms.

In order to appreciate some of the difficulties which are involved in the design of this project, imagine a discussion concerning the relative merits of automatic change over switch in the first place the specifications of the picture quality and staying. The discussion may then turn to the likelihood of faults developing in the sets. This is important not only because of the annoyance caused to the viewer by a failure but pay a higher initial cost of this device in return for an assurance that the extra cost will mean smaller maintenance costs. Therefore, from this little explanation, "Reliability can be defined as the characteristics of a component or of a system which may be expressed by the probability that it will perform a required function under started conditions for a specified period of time.

In this design and construction of this project, it is usually very important to note that maintainability is another area or aspect taken into consideration since high initial or production cost will lead to a low maintenance cost.

Therefore, maintainability of the probability that a device will be restored to operational effectiveness within a given period of time when the maintenance action as performance in accordance with prescribed procedures.

4.4 PROJECT EVALUATION AND TESTING

Considering the cost of this project one must note the fact that it is produced en mass. As such, to single handedly manufacture it without any industrial aided production machine will in no doubt reflect in the cost and the overall expenses, the initial market survey did not prove successful due to irregular price of item such as those used in this project.

First of all, before powering the system the multi-meter was set to the ohm(n) range, (xi) and check was made on all joints for possible short circuits. After the short circuit test the meter probes were run along wires and conducting lines to ensure that there are no open circuit.

Then the meter was set to voltage range (zero range) and the system was powered. The various power supply voltage rails were tested and confirmed.

All the supply was connected and powered; it was seen working satisfactorily.

4.5 APPLICATION OF THE PROJECT

1. This device could be used in that places where we have different sources of supply such as Mains from the grid, solar and generator.
2. This device could be used in industries for supplying the uninterruptable power supply to the industrial machines.
3. This device could be used in educational institutes and hospitals for supplying the uninterruptable power supply to the hospital or educational equipment.
4. This device is more compact and reliable as compared to the electrical ATS panels.
5. This device is less costly as compared to the other power control systems.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

In the “Power supply from four different sources: Solar, Inverter, Main and Generator” has been explained in this project with all its features and details. The significance of this project lies in its various advantages and wide places of applications such as Industries, Hospitals, Banks; etc. It has been developed by integrating Colleges/Schools, etc. It has been developed by integrating features of all the hardware components used. Presence of every module has been reasoned out and placed carefully thus contributing to the best working of the unit.

5.2 FUTURE SCOPE AND RECOMMENDATIONS

The Future Scope of the project is the addition of the supply for extra availability of the project and the convenience of the working operation would become easier if there is an extra source to the project. The higher specification of the components used would be a better idea for the durability of the project and project lies for the longer period. The main scope of this paper is to provide a continuous power supply to the output load through any of the source in the absence of any of the source. The paper can be further enhanced by using GSM, other sources like wind power and also then taking into consideration for using the best possible power whose cost remains lowest at that moment. The significance of this paper lies in its various advantages and wide places of applications where this project can be used efficiently.

5.3 REFERENCES

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